# An Introduction to Chirp and Stretch Processing

## 1 The transmitted chirp

The presentation writes the transmitted analytic signal as

$$p(t) = b(t)\exp(j2\pi F_c t), \tag{1}$$

where $F_c$ is the carrier frequency and the complex baseband waveform is

$$b(t) = \mathrm{rect}\left(\frac{t - \frac{1}{2}T_{\mathrm{pulse}}}{T_{\mathrm{pulse}}}\right)\exp\left\{j\left[\pi\gamma t^2 - 2\pi\left(\frac{B}{2}\right)t\right]\right\}, \tag{2}$$

with

$$\gamma = \frac{B}{T_{\mathrm{pulse}}}. \tag{3}$$

Here $t = 0$ is the start of transmission, $T_{\mathrm{pulse}}$ is the pulse duration, $B$ is the swept bandwidth, and $\gamma$ is the chirp rate in hertz per second. The rectangular factor confines the transmitted signal to $0 \le t \le T_{\mathrm{pulse}}$.

The instantaneous baseband frequency is the derivative of the phase divided by $2\pi$:

$$f_b(t) = \frac{1}{2\pi}\frac{d}{dt}\left(\pi\gamma t^2 - \pi B t\right) = \gamma t - \frac{B}{2}. \tag{4}$$

It therefore increases from $-B/2$ at $t = 0$ to $+B/2$ at $t = T_{\mathrm{pulse}}$. The corresponding RF frequency increases from $F_c - B/2$ to $F_c + B/2$.

Apart from an irrelevant constant phase, the baseband signal may also be written as a chirp centered at $T_{\mathrm{pulse}}/2$:

$$b(t) \doteq \mathrm{rect}\left(\frac{t - \frac{1}{2}T_{\mathrm{pulse}}}{T_{\mathrm{pulse}}}\right)\exp\left[j\pi\gamma\left(t - \frac{T_{\mathrm{pulse}}}{2}\right)^2\right], \tag{5}$$

where $\doteq$ denotes equality up to a constant complex phase.

For a single point scatterer with propagation delay $T_{\mathrm{prop}}$, the idealized received signal is a delayed and scaled copy of the transmitted signal:

$$p_{rx}(t) = \alpha p(t - T_{\mathrm{prop}}). \tag{6}$$

The complex number $\alpha$ accounts for reflectivity, propagation loss, and other amplitude and phase effects that are not central to the present discussion.

## 2 What range compression means

Before processing, the response from a small point scatterer lasts roughly $T_{\text{pulse}}$, because it is a delayed copy of the long transmitted chirp. Range compression uses knowledge of the transmitted waveform to replace this long response by a narrow correlation peak at the scatterer's propagation delay.

After carrier downconversion, the received baseband signal is

$$r(t) = p_{rx}(t)\exp(-j2\pi F_c t) = \alpha\exp(-j2\pi F_c T_{\text{prop}})b(t - T_{\text{prop}}). \tag{7}$$

The factor $\exp(-j2\pi F_c t)$ performs carrier downconversion; it is not motion compensation.

The matched filter has impulse response

$$h(t) = b^*(-t). \tag{8}$$

Convolving $r(t)$ with $h(t)$ is equivalent to correlating the received signal with the transmitted baseband waveform:

$$s(\tau) = \int_{-\infty}^{\infty} r(t)b^*(t-\tau)\,dt. \tag{9}$$

Define the autocorrelation of $b(t)$ by

$$R_b(\Delta) = \int_{-\infty}^{\infty} b(t)b^*(t-\Delta)\,dt. \tag{10}$$

Substitution of (7) into (9) gives

$$s(\tau) = \alpha\exp(-j2\pi F_c T_{\text{prop}})R_b(\tau - T_{\text{prop}}). \tag{11}$$

Thus the user's interpretation is correct: the matched filter changes the response of a point scatterer from a waveform having duration approximately $T_{\text{pulse}}$ into the autocorrelation of the baseband waveform. The autocorrelation has a narrow central peak when $b(t)$ has a large bandwidth. The output is not literally a Dirac delta, but it is often represented as one when its width is negligible on the scale of interest.

For the rectangularly windowed linear-FM waveform in (2), the magnitude of the exact autocorrelation has the form

$$|R_b(\Delta)| = \begin{cases} (T_{\text{pulse}} - |\Delta|)\,|\text{sinc}[\gamma\Delta\,(T_{\text{pulse}} - |\Delta|)]|, & |\Delta| < T_{\text{pulse}}, \\ 0, & |\Delta| \geq T_{\text{pulse}}, \end{cases} \tag{12}$$

where

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}. \tag{13}$$

For a large time-bandwidth product $BT_{\text{pulse}} \gg 1$, the central peak has a characteristic width of approximately

$$\Delta\tau \simeq \frac{1}{B}. \tag{14}$$

The approximate pulse-compression ratio is consequently

$$\frac{T_{\text{pulse}}}{\Delta\tau} \simeq BT_{\text{pulse}}. \tag{15}$$

This permits a long pulse, which carries substantial energy, to provide the delay resolution associated with a short pulse of duration about $1/B$.

In a monostatic radar, $T_{\text{prop}} = 2R/c$, so the delay resolution corresponds approximately to

$$\Delta R \simeq \frac{c}{2B}. \tag{16}$$

In a bistatic radar, $cT_{\text{prop}}$ is the sum of the transmitter-to-scatterer and scatterer-to-receiver distances. A compressed delay bin therefore represents a constant-bistatic-range surface.

## 3 The same operation in the frequency domain

Use the Fourier-transform convention

$$X(f) = \int_{-\infty}^{\infty} x(t) \exp(-j2\pi ft)\, dt, \tag{17}$$

and denote the transform of $b(t)$ by $\mathcal{B}(f)$. The transform of the received baseband signal is

$$R(f) = \alpha \exp(-j2\pi F_c T_{\text{prop}}) \mathcal{B}(f) \exp(-j2\pi f T_{\text{prop}}). \tag{18}$$

Matched filtering is multiplication by the conjugate spectrum:

$$S(f) = R(f)\mathcal{B}^*(f) \tag{19}$$

$$= \alpha \exp[-j2\pi(F_c + f)T_{\text{prop}}]|\mathcal{B}(f)|^2. \tag{20}$$

The inverse Fourier transform of $|\mathcal{B}(f)|^2$ is precisely the autocorrelation $R_b(\tau)$. Equations (11) and (20) therefore describe the same matched-filter operation in the time and frequency domains.

The slides use the approximation

$$|\mathcal{B}(f)|^2 \approx C \ \text{rect}\left(\frac{f}{B}\right), \tag{21}$$

where $C$ is an unimportant scale factor. With that approximation,

$$S(f) \propto \exp[-j2\pi(F_c + f)T_{\text{prop}}] \,\text{rect}\left(\frac{f}{B}\right). \tag{22}$$

An inverse transform produces a shifted sinc-like peak whose characteristic width is $1/B$.

## 4 Finite pulse duration versus finite bandwidth

There is an important mathematical qualification to (21). A nonzero waveform cannot be both strictly time limited and strictly band limited. In particular:

- The rectangular factor in (2) makes $b(t)$ exactly zero outside a finite time interval.
- Its Fourier transform therefore cannot also be exactly zero outside the interval $[-B/2, B/2]$. Abruptly turning the pulse on and off produces spectral sidelobes extending to arbitrarily large frequencies.
- Conversely, an exactly rectangular spectrum has an infinite-duration sinc function as its inverse Fourier transform and cannot represent a strictly finite pulse.

Thus $B$ in this model is the chirp's swept or effective bandwidth, not an exact compact spectral support. When $BT_{\text{pulse}}$ is large, most of the energy lies in the swept band and the spectrum is approximately flat there. Treating $|\mathcal{B}(f)|^2$ as a rectangle is then a useful idealized model, but it suppresses the actual transition regions and spectral sidelobes.

This distinction also explains two related descriptions of the compressed pulse:

- Under the ideal rectangular-spectrum approximation, the compressed response is an infinite-duration sinc function.
- For the exactly time-limited chirp, the exact autocorrelation is zero outside $|\Delta| < T_{\text{pulse}}$, as in (12), but has a narrow central peak and sidelobes inside that interval.

Both models predict a main-peak scale of about $1/B$ when the time-bandwidth product is large. Window tapering can reduce spectral or range sidelobes, but it broadens the main correlation peak; this is the usual resolution–sidelobe tradeoff.

## 5 Chirp-mode processing

In chirp mode the receiver digitizes the broadband echo and performs matched filtering. An idealized processing chain is

$$p_{rx}(t) \xrightarrow{\times \exp(-j2\pi F_c t)} r(t) \xrightarrow{\mathcal{F}} R(f) \xrightarrow{\times \mathcal{B}^*(f)} S(f). \tag{23}$$

The first multiplication is carrier downconversion. The multiplication by $\mathcal{B}^*(f)$ is the matched filter and performs range compression.

The presentation decomposes the propagation delay as

$$T_{\text{prop}} = T_{\text{scp}} + T_{\text{sct}} + T_\epsilon, \tag{24}$$

where $T_{\text{scp}}$ is the scene-center propagation term, $T_{\text{sct}}$ is the scatterer's delay relative to the scene center, and $T_\epsilon$ is the flat-wavefront approximation error. Chirp-mode motion compensation multiplies the range-compressed data by

$$G_{mc}(f) = \exp[+j2\pi(F_c + f)T_{\text{scp}}] \tag{25}$$

$$= \underbrace{\exp(+j2\pi F_c T_{\text{scp}})}_{extslow-timecompensation} \underbrace{\exp(+j2\pi f T_{\text{scp}})}_{extfast-timecompensation}. \tag{26}$$

This removes the scene-center propagation phase and leaves

$$S_{mc}(f) \propto \exp[-j2\pi(F_c + f)(T_{\text{sct}} + T_\epsilon)] \operatorname{rect}\left(\frac{f}{B}\right). \tag{27}$$

## 6 Stretch-mode processing

Stretch mode transmits the same LFM waveform but performs the receiver's range processing differently. The received chirp is multiplied by the conjugate of a reference chirp. If the reference delay is $T_{\text{ref}}$, the product is schematically

$$p(t - T_{\text{prop}})p^*(t - T_{\text{ref}}). \tag{28}$$

The two quadratic chirp phases cancel, leaving a nearly sinusoidal beat signal whose frequency is, up to the sign convention,

$$f_{\text{beat}} = -\gamma(T_{\text{prop}} - T_{\text{ref}}). \tag{29}$$

Thus propagation delay is mapped one-to-one into beat frequency. A Fourier transform of the stretch data separates different beat frequencies into different range bins. Since an observation of duration $T_{\text{pulse}}$ resolves beat frequencies on a scale of approximately $1/T_{\text{pulse}}$, the corresponding delay resolution is

$$\Delta T \simeq \frac{1/T_{\text{pulse}}}{|\gamma|} = \frac{1}{B}. \tag{30}$$

Stretch processing therefore obtains the same fundamental range resolution as the matched-filter method. Its practical advantage is that the dechirped signal can occupy a much smaller receiver bandwidth when only a limited range swath is of interest.

In the idealized notation of the presentation, the range-compressed and motion-compensated stretch result is

$$s_{mc,\text{stretch}}(t) = \exp[-j2\pi(F_c + \gamma t)(T_{\text{sct}} + T_\epsilon)]\,\text{rect}\left(\frac{t}{T_{\text{pulse}}}\right). \tag{31}$$

Equation (31) is an output expression, not the factor by which the unprocessed received signal is multiplied. The corresponding idealized scene-center motion-compensation factor is

$$G_{mc,\text{stretch}}(t) = \exp[+j2\pi(F_c + \gamma t)T_{\text{scp}}]. \tag{32}$$

In a practical stretch receiver, an appropriately delayed and phased reference chirp can incorporate some or all of this compensation directly into the dechirping operation.

Using the correspondence

$$f = \gamma t, \qquad |\gamma| T_{\text{pulse}} = B, \tag{33}$$

equations (27) and (31) have the same mathematical form. For a negative-slope chirp, the sign reverses the frequency ordering, but the rectangular window is unchanged because it is an even function.

# 7 Summary

1. Chirp and stretch modes can transmit the same finite-duration LFM signal $p(t)$.

2. In chirp mode, range compression is performed by a matched filter. The response of a point scatterer becomes the autocorrelation of the baseband waveform, centered at the propagation delay.

3. The correlation peak has characteristic width $1/B$, rather than $T_{\text{pulse}}$, giving an approximate compression ratio $BT_{\text{pulse}}$.

4. Stretch mode instead dechirps the received signal with a reference chirp, mapping propagation delay into beat frequency. A subsequent Fourier transform separates ranges, again with delay resolution about $1/B$.

5. A pulse cannot be both exactly time limited and exactly band limited. The rectangular spectrum used in the slides is an engineering approximation to the spectrum of a finite-duration, large time-bandwidth-product chirp.